Project: Worked examples
Title: Fatigue damage (Palmgren–Miner)
Status: DRAFT · Author: TempestOS (2026-10-23) · Checker: — (—) · Issued: —

## 1. Scope and method

Cumulative fatigue damage of a stress-range spectrum against an S–N curve of one slope through a reference point, with an optional endurance limit below which a range does no damage; the spectrum repetitions to failure.

| WHAT IT CALCULATES | RELATION |
|---|---|
| Cycles to failure at each stress range, from the S–N curve (unbounded at or below the endurance limit S_e) | N_i = N_ref·(S_ref/S_i)^m |
| Damage done by each block of cycles | d_i = n_i/N_i |
| Total damage, and how many repetitions of the spectrum the part survives | D = Σ d_i; repetitions = 1/D |
| Acceptance by the Palmgren–Miner rule | D ≤ 1 |

Basis: Fatigue resistance (S–N curves by detail category)[1](§8); Cumulative fatigue damage[2](§6-15).

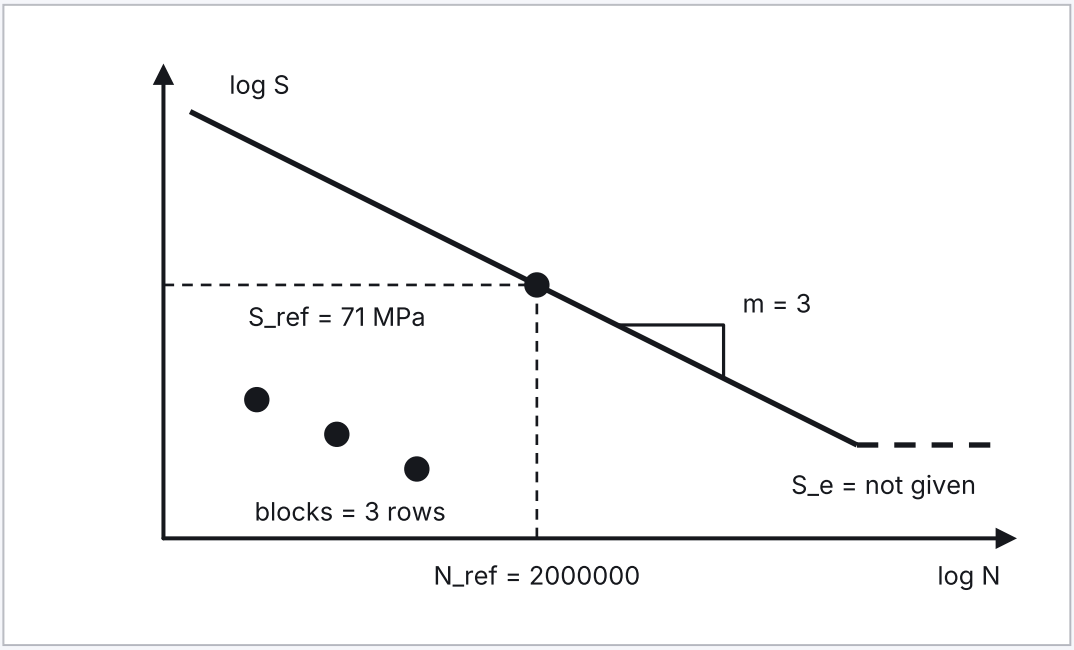



Figure 1 — A single-slope S–N line on log axes through its reference point, the endurance limit as a cut-off, and the stress-range blocks as listed (three drawn, representatively). Not to scale; the labels are the inputs as entered.

## 2. Inputs

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| S_ref | Reference stress range (detail category or material fatigue strength) | 71 | MPa | spec: SPEC-MECH-03 design criteria | 1 … 2000 MPa |

| SYMBOL | DESCRIPTION | VALUE | UNIT | SOURCE | VALID RANGE |
|---|---|---|---|---|---|
| N_ref | Cycles at the reference stress range | 2000000 | | spec: SPEC-MECH-03 design criteria | 10000 … 1000000000 |
| m | S–N slope | 3 | | spec: SPEC-MECH-03 design criteria | 2 … 15 |
| blocks | Stress-range blocks | 3 row(s) | | spec: LOAD-SCHEDULE-01 | any |

| TABLE ROW | VALUES |
|---|---|
| blocks[1] | S = 100 MPa; n = 100000 |
| blocks[2] | S = 60 MPa; n = 500000 |
| blocks[3] | S = 40 MPa; n = 2000000 |

## 3. Assumptions

A1. Damage accumulates linearly and independently of the order of the blocks. (Palmgren–Miner rule.)
A2. One S–N slope over the whole range; no second slope below a knee. (Simplified curve; a two-slope curve is entered as two sheets or a lower slope.)

## 4. Working

[4] N_1 = N_ref · (S_ref / S_1)^m [1](§8)
= 2000000 × (71 MPa / 100 MPa)^3
= 715820
Life at block 1

[5] d_1 = n_1 / N_1
= 100000 / 715820
= 0.1397
Damage of block 1

[6] N_2 = N_ref · (S_ref / S_2)^m [1](§8)
= 2000000 × (71 MPa / 60 MPa)^3
= 3314000
Life at block 2

[7] d_2 = n_2 / N_2
= 500000 / 3314000
= 0.15088
Damage of block 2

[8] N_3 = N_ref · (S_ref / S_3)^m [1](§8)
= 2000000 × (71 MPa / 40 MPa)^3
= 11185000
Life at block 3

[9] d_3 = n_3 / N_3
= 2000000 / 11185000
= 0.17882
Damage of block 3

[10] D = Σ d_i [2](§6-15)
= 0.1397 + 0.15088 + 0.17882
= 0.46939
Total damage (Palmgren–Miner)

[11] R_f = 1 / D
= 1 / 0.46939
= 2.1304
Spectrum repetitions to failure

## 5. Results

| SYMBOL | VALUE | DESCRIPTION |
|---|---|---|
| D | 0.46939 | Total damage |
| R_f | 2.1304 | Repetitions to failure |
| N_1 | 715820 | Life at block 1 |
| d_1 | 0.1397 | Damage of block 1 |
| N_2 | 3314000 | Life at block 2 |
| d_2 | 0.15088 | Damage of block 2 |
| N_3 | 11185000 | Life at block 3 |
| d_3 | 0.17882 | Damage of block 3 |

## 6. Checks (governing first)

| RESULT | CHECK | DEMAND | CAPACITY | UTILISATION | REF |
|---|---|---|---|---|---|
| PASS | Cumulative damage D ≤ 1 | D = 0.46939 | D ≤ 1 | 0.469 | [2](§6-15) |

## 7. Applicability

No warnings.

## 8. Verification

Template mech.fatigue-miner@1.0.1 Content hash 82e9e2cff5d85046 Engine calc-engine/1.0.0
Known-answer cases passed 2/2 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

## 9. References

1. British Standards Institution. Eurocode 3: Design of Steel Structures—Part 1-9: Fatigue. BS EN 1993-1-9. BSI.
2. Budynas RG, Nisbett JK. Shigley's Mechanical Engineering Design. 11th ed. McGraw-Hill Education; 2020.